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9th International Mathematics Assessments for Schools (2019-2020)

Junior Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Junior Division Round 2

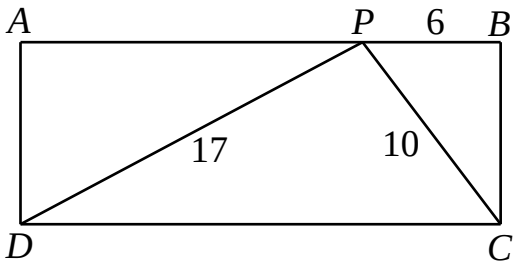
Questions 1 to 5, 4 marks each

1. What is the units digit of the expression: $(-2)^{2019}+(-2)^{2020}$?

（A）0　　（B）2　　（C）4　　（D）6　　（E）8

Answer：_____

2. In the figure below, point P lies on rectangle $ABCD$ such that $PB=6$ cm, $PC=10$ cm and $PD=17$ cm. What is the area, in cm^2, of rectangle $ABCD$?



（A）84　　（B）168　　（C）170　　（D）230　　（E）272

Answer：_____

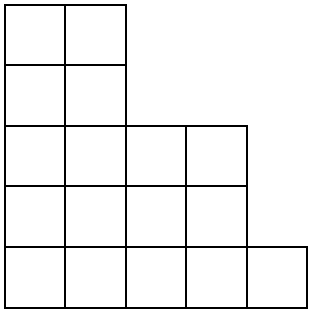
3. If each of the variables x, y, and z is increased by 1 at the same time, then which one of the following algebraic expressions will remain the same value?

（A）$x^2+y^2+z^2$　　（B）$x^3+y^3+z^3-3xyz$　　（C）$xy^2+yz^2+zx^2$

（D）$2x^2-2xy+2y^2-xz-yz+z^2$　　（E）$3x^2-4xy+3y^2-2xz-2yz+2z^2$

Answer：_____

4. In the figure below, color five unit squares black such that any two black unit squares are not located in the same row or column. How many ways can we color the figure?



（A）4　　（B）8　　（C）12　　（D）24　　（E）120

Answer：_____

5. How many ordered pairs of positive integer (x, y) satisfy $x^2 + 5y^2 = 2020$?

（A）0　　（B）1　　（C）2　　（D）3　　（E）4

Answer：_____

Questions 6 to 13, 5 marks each

6. There are four consecutive positive integers, three of which have a product of 44030. What is the fourth positive integer?

Answer：_____

7. There are four rectangular pieces with dimensions of $1\text{ cm} \times 2\text{ cm}$, $3\text{ cm} \times 4\text{ cm}$, $5\text{ cm} \times 6\text{ cm}$ and $7\text{ cm} \times 8\text{ cm}$ respectively. They are to be placed on a larger rectangular graphing paper without overlapping. The sides of each piece should be parallel to the edges of the graph paper and that one vertex from each of the four pieces should coincide at a point (that is, this point is also the vertex of four rectangles). What is the smallest possible area, in cm^2, of the graphing paper?

Answer：_____ cm^2

8. Select two different numbers from the set of numbers {1, 2, ..., 10} such that they are not relatively prime. How many different pairs of numbers are there in total? (Note that selecting numbers *a* and *b* is the same as selecting *b* and *a*.)

Answer：_____ pairs

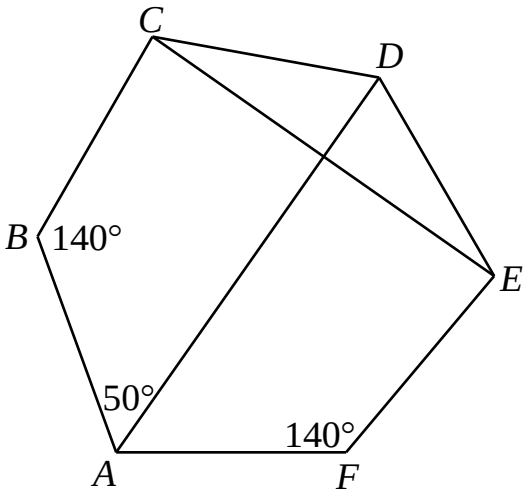
9. If $x^3 + 2x^2 + 3x + 4 = a(x-4)^3 + b(x-4)^2 + c(x-4) + d$ is true for any value of *x*, then what is the value of the constant *c*?

Answer：_____

10. The sum of three non-negative integers *a*, *b*, and *c* is 40, where *a* is always an odd number. How many ordered triples (*a*, *b*, *c*) will satisfy the condition?

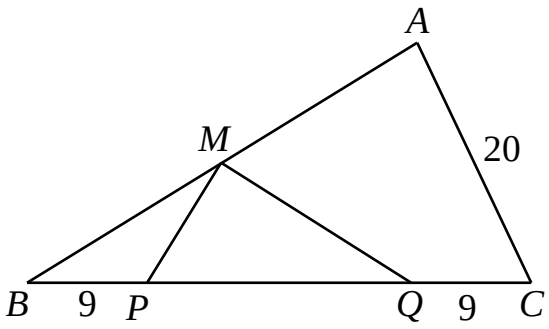
Answer：_____

11. In the figure below, the side lengths of hexagon *ABCDEF* are all equal, $\angle ABC = \angle AFE = 140°$ and $\angle BAD = 50°$.
Find the measure, in degrees, of $\angle BCE$.



Answer : _____ °

12. In the figure below, points *P* and *Q* lie on side *BC* of triangle *ABC* and point *M* is the midpoint of side *AB*. If $BP = CQ = 9$ cm, $AC = 20$ cm and $PM \perp MQ$, then what is the length, in cm, of side *BC*?



Answer : _____ cm

13. In the algebraic expression $a^b + c^d + e^f + g^h$, replace each letter with the numbers 1, 2, 3, 4, 5, 6, 7, and 8, such that each letter is represented by a different number. What is the smallest possible value of the said expression?

Answer : _____

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. A three-digit number $\overline{abc}$ is a prime number, such that a, b and c represent different digits, and that $a+b+c$ is an odd number. Prove that $a+b+c$ is also a prime number.

15. In the quadrilateral *ABCD*, *AB*//*CD*, point *E* is the intersection of diagonal lines *AC* and *BD*, and point *F* is the midpoint of *AD*. Extend *EF* to intersect *BC* at point *G* and let A_1, B_1, C_1 and D_1 be the feet of perpendiculars from *A*, *B*, *C* and *D* to line *FG*, respectively.

（1）Prove $\frac{AA_1}{CC_1}=\frac{BB_1}{DD_1}$；(5 marks)

（2）Prove $FE \geq GE$。(15 marks)

